

Factor trinomials of the form $x^2 + bx + c$

Algebra I | CK-12: Factoring when the Leading Coefficient Equals 1

Key Idea

To factor $x^2 + bx + c$, find two numbers that **multiply to** c and **add to** b . Those numbers go into $(x + _)(x + _)$.

Key Terms

- **Trinomial** — a three-term expression
- **Product-sum** — multiply to c , add to b

Worked steps

Step 1: List factor pairs of c .

Step 2: Pick the pair that adds to b ; place them in $(x + _)(x + _)$.

Example: $x^2 + 5x + 6$: $2 \cdot 3 = 6$ and $2 + 3 = 5$, so $x^2 + 5x + 6 = (x + 2)(x + 3)$.

You Try

Factor $x^2 + 7x + 10$.